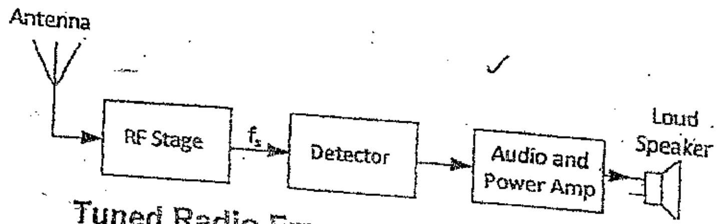
EE 435

RADIO RECEIVERS

Of all the radio receivers proposed at one time or the other, only two of them are of commercial significance. These are the Tuned Radio Frequency receiver and the Super-heterodyne receiver.

The Tuned Radio Frequency Receiver

- It is the most elementary receiver design consisting primarily of RF amplifier, a detector, Audio and Power amplifier stages. The tuned radio frequency (TRF) receiver has three stages of RF amplification, with each stage preceded by a separate variable tuned circuit.



Tuned Radio Frequency Receiver (TRF)

Sensitivity and Selectivity: Two major characteristics of any radio receiver are its sensitivity and selectivity.

- Sensitivity of a radio receiver can be defined as the minimum input RF signal to the receiver capable of producing a specified audio signal at the output. The range of sensitivities for communication receivers vary from the millivolt (10^{-3} volts) region for low cost AM receivers, down to the nanovolt (10^{-9} volts) region, for ultra-sophisticated units for more exacting applications.
- Selectivity of a radio receiver is defined as the extent to which a receiver is capable of differentiating between the desired signal and other frequency signals (unwanted radio signal and noise).

Problem of TRF Selectivity: A standard AM receiver can handle a signal up to a maximum of 5KHz, which subsequently generates upper and lower sidebands extending 5KHz above and below the carrier frequency. The total signal has a 10KHz bandwidth. Therefore the optimum receiver selectivity is thus 10KHz. However if a 5KHz bandwidth were selected, the upper and lower sidebands would only extend 2.5KHz above and below the carrier. The radio would lack full possible reception. On the other hand an excessive selectivity of 20KHz bandwidth would result in reception of unwanted...

adjacent radio signals and additional external noise that is directly proportional to the bandwidth selected. Unfortunately, TRF receivers suffer from problems of poor selectivity.

Estimating TRF Selectivity: Let us consider a standard AM broadcast band receiver that spans the frequency range from 550 to 1550 kHz. If the approximate center of this frequency range of 1000 kHz is considered, then

$$Q = \frac{f_r}{\mathrm{BW}} = \frac{1000\ \mathrm{kHz}}{10\ \mathrm{kHz}} = 100$$

Note that Q is called the quality factor of the filter and it is the measure of how selective (narrow) its passband is compared to its center frequency, f_r.
Now, since the Q of the tuned circuit remains fairly constant as its capacitance is varied, a change to 1550 kHz will change the BW to 15.5 kHz.

$$Q = \frac{f_r}{\mathrm{BW}} \quad \therefore \quad \mathrm{BW} = \frac{f_r}{Q} = \frac{1550\ \mathrm{kHz}}{1000} = 15.5\ \mathrm{kHz}$$

The receiver's BW is now too large, and it will suffer from adjacent station interference and increase noise.
On the other hand, the opposite problem is encountered at the lower end of the frequency range. At 550 kHz, the BW is given by,

$$\mathrm{BW} = \frac{f_r}{Q} = \frac{550\ \mathrm{kHz}}{100} = 5.5\ \mathrm{kHz}$$

The fidelity of the reception is now impaired. The maximum intelligence frequency possible is 5.5 kHz/2 or 2.75 kHz.

Exercise

A TRF receiver is to be designed with a single tuned circuit using a 10 μH inductor.
(a) Calculate the capacitance range of the variable capacitor required to tune from 550 to 1550 kHz.
(b) The ideal 10 kHz BW is to occur at 1100 kHz. Determine the required Q.
(c) Calculate the BW of this receiver at 550 kHz and 1550 kHz.
Hint use the formula to calculate C

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

AM Detection: If the AM signal is passed through a nonlinear device, difference frequencies between the carrier and sidebands will be generated and these frequencies are, in fact the intelligence

Detection of amplitude modulated signals requires a nonlinear electrical network. An ideal nonlinear curve for this is one that affects the positive half-cycle of the modulated wave differently than the negative half-cycles. This distorts an applied voltage wave of zero average value so that the average resultant current varies with the intelligence signal amplitude. The curve shown in Figure 19(a) is called an *ideal curve* because it is linear on each side of the operating point P and does not introduce harmonic frequencies.

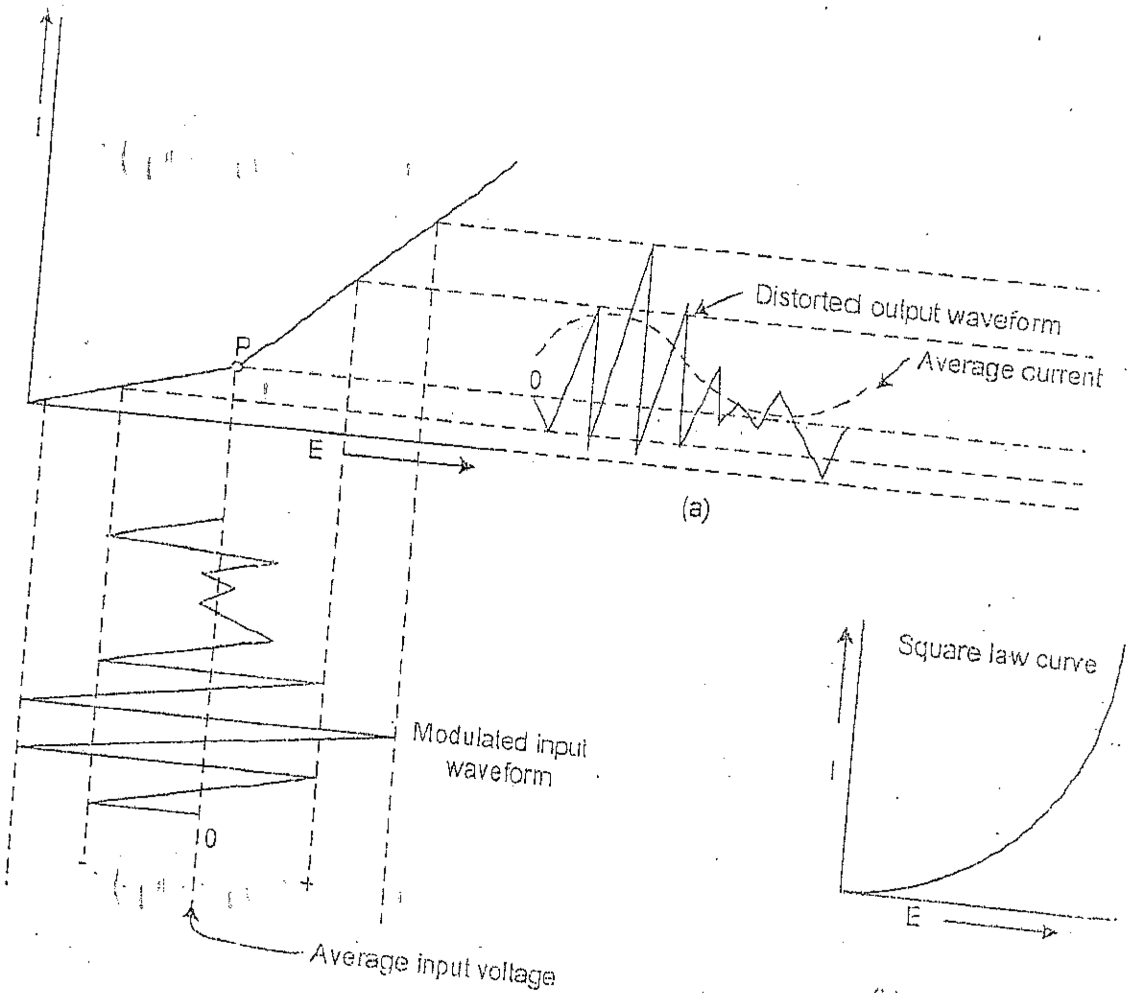


Figure 19: Nonlinear Device used as Detector — Ideal Curve

When the input to an ideal nonlinear device is a carrier and its sidebands, the output contains the following frequencies:

1. The carrier frequency
2. The upper sideband — the original components
3. The lower sideband
4. A dc component
5. A frequency equal to the carrier minus the lower sideband and the upper sideband minus the carrier, which is the original signal frequency.

The detector reproduces the signal frequency by producing a distortion of a desirable kind in its output. When the output of the detector is impressed upon a low-pass filter, the radio frequencies are suppressed and only the low-frequency intelligence signal and dc components are left. This is shown as the dashed average current curve in Figure 19(a). In some practical detector circuits, the nearest approach to the ideal curve is the square-law curve shown in Figure 19(b).

The Super-heterodyne Radio Receiver

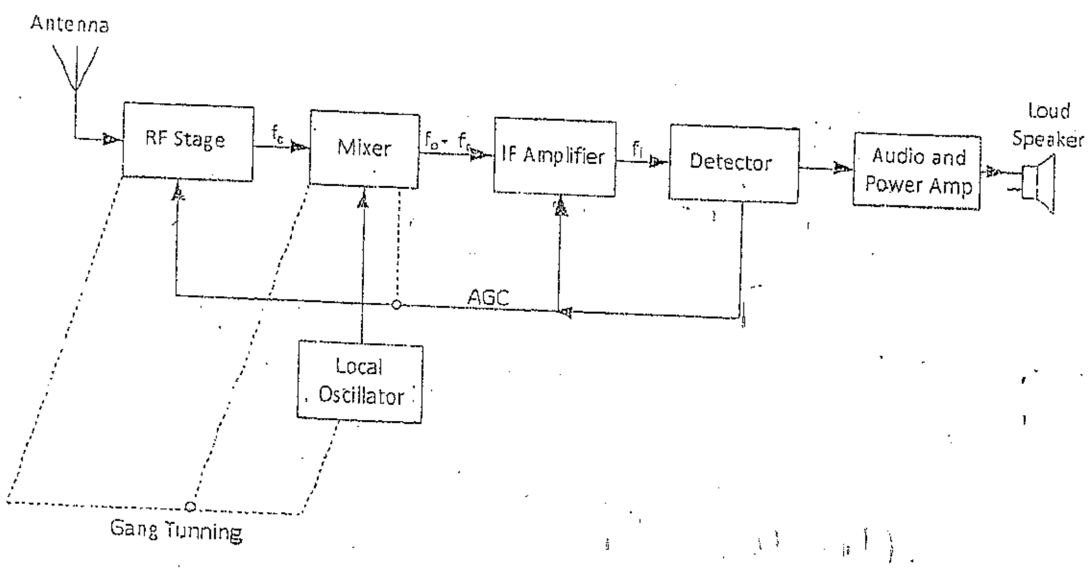


Figure 20: Superheterodyne Radio Receiver

The block diagram of a superheterodyne radio receiver is shown in figure above. In the superheterodyne receiver, the signal voltage is combined with the local oscillator voltage and normally converted into a signal of lower fixed frequency. This signal at the intermediate frequency contains the same modulation as the original carrier, and it is amplified and detected to reproduce the original information. The superheterodyr thus has the same essential components as the TRF receiver in addition to the mixe local oscillator and intermediate frequency (IF) amplifier. A dc level proportional to th received signal's strength is extracted from the detector stage and fed back to the I amplifiers and sometimes to the mixer and/or the RF amplifier. This is the automati gain control (AGC) level, which allows relatively constant receiver output for widel variable radio signals.

A constant frequency difference is maintained between the local oscillator and the RF circuits, normally through capacitance tuning in which all the capacitors are ganged together and operated in unison by one control knob. The IF amplifier provides most of the gain (sensitivity) and bandwidth requirements of the receiver; that is, it provides the bulk of radio-frequency signal amplification at a fixed frequency. This allows for constant bandwidth over the entire band of the receiver and is the key to superior selectivity of the superheterodyne receiver over the TRF receiver. Selectivity and sensitivity of the superhet are usually fairly uniform throughout its tuning range and is not subjected to the variations that beset the TRF receivers. The RF circuits are now used mainly to select wanted frequency, to reject interference such as image frequency, and to reduce the noise-figure of the receiver especially at high frequencies.

Reason for use and functions of RF amplifier: A receiver having an RF stage is undoubtedly superior in performance to the receiver without one, all being equal. On the other hand, there are some instances in which an RF amplifier is uneconomical, that is, where its inclusion would increase the cost of the receiver significantly while improving performance marginally.

The advantages of having RF amplifier are as follows:

1. Greater gain, that is, better sensitivity.
2. Improved image frequency rejection.
3. Improved signal-to-noise ratio.
4. Improved rejection of adjacent unwanted signals, that is, better selectivity.
5. Better coupling of the receiver to the antenna (important at VHF and above).

Image Frequency and its Rejection: The local oscillator frequency is usually made higher than the station or signal frequency, f_s by an amount equivalent to the intermediate frequency (IF). That is, $f_o - f_s = f_i$. Then if another frequency f_{si}, is picked by the receiver, the mixer circuit will together with the local oscillator frequency f_o produce a difference frequency f_i' which resembles the original intermediate frequency f_i . The signal frequency f_{si} that causes the production of the frequency f_i' is called image frequency. Thus image frequency is the unwanted signal frequency which if allowed to enter the radio receiver would cause interference.

Image frequency rejection is the ability of the radio receiver to put off the unwanted signal frequency outside its operating band, so as not to interfere with the operating station frequency. The tune circuit of the mixer makes the rejection very possible because of its particular characteristics. A worked example will illustrate it better.

Example: Determine the image frequency for a standard broadcast band radio using a 455 kHz IF which is tuned to a 650 kHz station. Show whether image frequency rejection will be possible through your results and the tuned circuit characteristics.

Solution: Image frequency is given by:

$f_{si} = f_s + 2f_i$; But f_s = 650 kHz (Station frequency).

For 455 kHz IF, f_{si} = 650 + 2 x 455 = 1560 kHz.

But the allowable band for station frequency of 650 kHz = $2f_s$ = 2 x 650 kHz = 1300 kHz.

Within the band, there will be interference. Outside the band, there will be no interference. Thus for the 455 kHz IF, image frequency rejection is possible since the image frequency is outside the operating band of the station. This is shown below by the mixer tuned circuit characteristics.

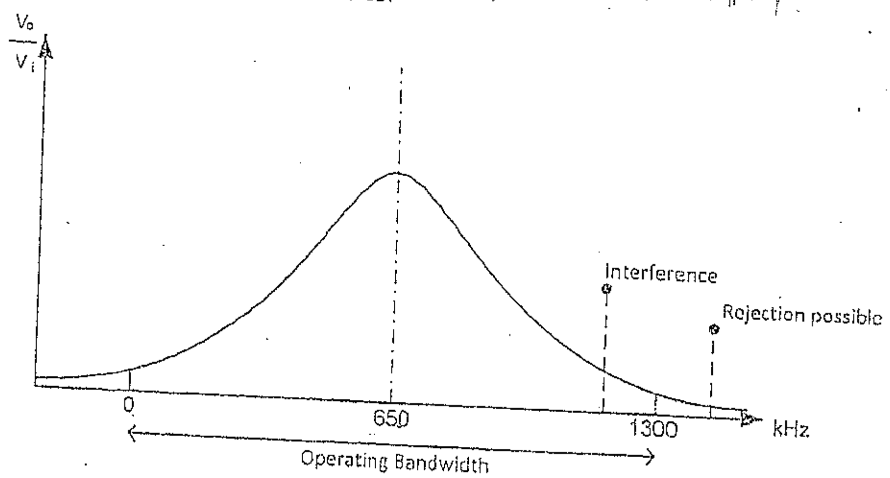


Figure 21: Mixer Tuned Circuit Characteristics

Exercise: Assign 2A

Determine the image frequency for a standard broadcast band radio using 450 kHz IF with another using 250 kHz IF. Both are tuned to a 700 kHz station. Determine the station bandwidth. State whether image frequency rejection is possible in both cases. Use the mixer tuned circuit xteristics and your result to illustrate your point.

TELEVISION

Introduction: The word "Television" is derived from two words, "Tele" meanir 'distance' and "Vision" meaning 'seeing'. Hence Television actually means 'seeing at distance'. The concept of television was developed in the 1920s. Commerci broadcasting started in the 1940s. In Nigeria it started in the late 1950s in the presei South West zone, anchored by the late Chief Obafemi Awolowo.

Transmitter Principles

A television transmitter is made up of two separate transmitters – the ~~sound~~ Sound transmitter and the video or picture transmitter. The sound transmitter is an FM system similar to broadcast FM radio; while the video or picture transmitter utilises the AM technique. During transmission the composite transmitted signal is a combination both AM and FM principles. This is done to minimize interference effects between the two at the receiver because FM receiver is relatively insensitive to amplitude modulation and an AM receiver has rejection capabilities to frequency modulation.

Fm is transmitted in frequency modulation
Video is transmitted in amplitude modulation

The Simplified TV System

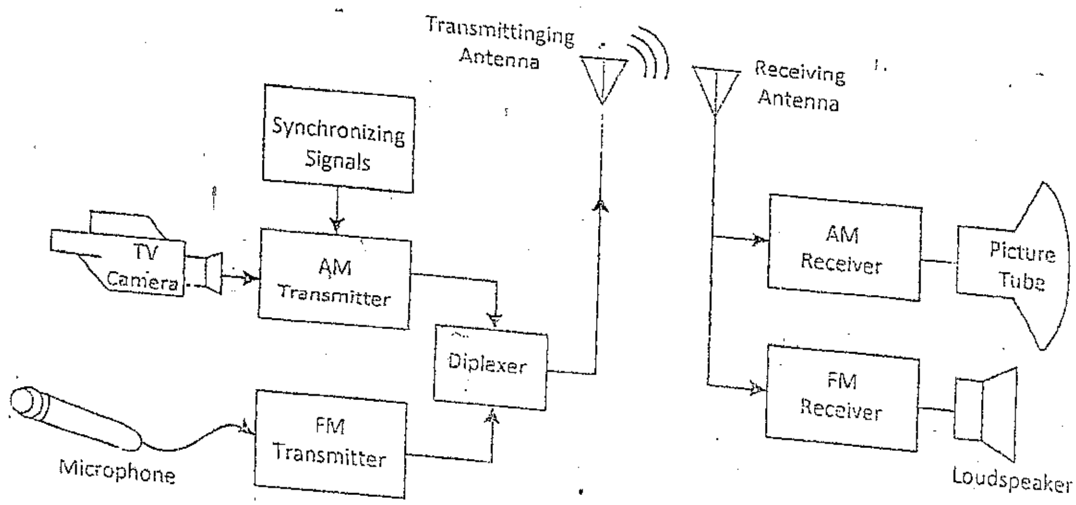


Figure 22: Simplified Television System

Figure 22 shows a simplified block diagram of a TV system. The TV camera convert the picture scene into an electrical signal. Thus the camera is thus a transducer t converts light energy into electrical energy. At the receiver, the CRT picture tube another transducer that converts the electrical energy back into light energy. microphone and the loudspeaker are similarly related transducers for soı transmission and reception. The transmitting antenna converts the electrical energy into electromagnetic energy required for transmission through space, while the receiving antenna converts the electromagnetic energy into electrical energy for the receiver to process into light energy.

The diplexer is feeding the transmitting antenna with both video and sound signals while not allowing either to be fed back into one another's transmitter.

Scanning

The most widely used pickup device in TV cameras is the *charge couple device* (CCD). The CCD is a solid state chip consisting of thousands or millions of photosensitive cells arranged in a two-dimensional array. When light (photons) strike the CCD surface, the light information is converted to an electronic analog of the light.

Scanning is like reading a book. It starts from the upper left hand corner of the screen and moves to the right hand side of the screen tilting slightly downwards to the end of the line. The scene to be televised is focused onto the photosensitive cells in the CCD imaging device. For simplicity sake for example the letter "T" is picked. Instead of millions of cells, let it be assumed that the system has 30 cells arranged in 6 rows and 5 columns. Each separate cell is called a *pixel* meaning "picture element". The greater the number of pixels, the better the quality (or resolution) of the transmitted picture.

As the scene letter "T" is focused on the light-sensitive area so that all of rows 1 and 6 are illuminated, while all of row 2 is dark and the centers of rows 3, 4 and 5 are dark.

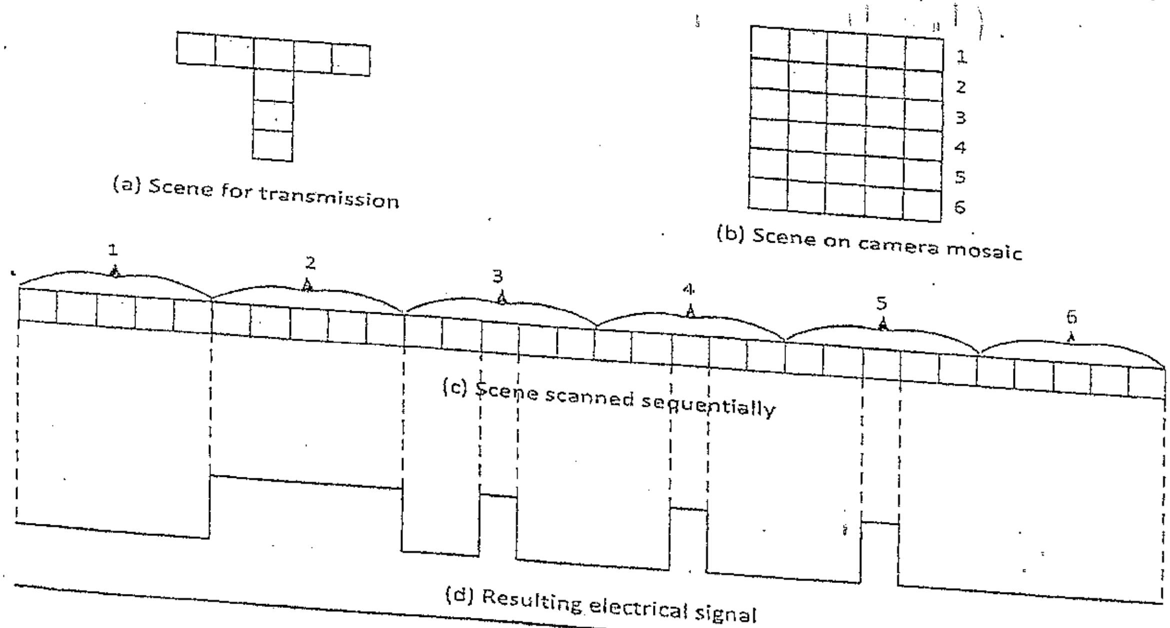


Figure 23: Simplified Scanning representation

Now if the rows of the scene are scanned sequentially and if the retrace interval is essentially zero, then Figure 23(c) shows the sequential breakup of information. The *Retrace Interval* is the time it takes an electron beam to move from the end of one line to the start of the next lower line. It is usually accomplished very rapidly. The variable light on the photosensitive cells results in a similar variable voltage being developed at the CCD's output, as shown in Figure 23(d).

Interlaced Scanning

The *frame frequency* is the number of times per second that a complete picture scene is traced. The rate for broadcast TV is 30 times per second. In other words, a complete scene (frame) is traced every $\frac{1}{30}$ s. Thirty frames per second is not enough to keep the human eye from perceiving *flicker* as a result of non-continuous visual presentation. If the frame frequency were increased to 60 frames per second, the flicker

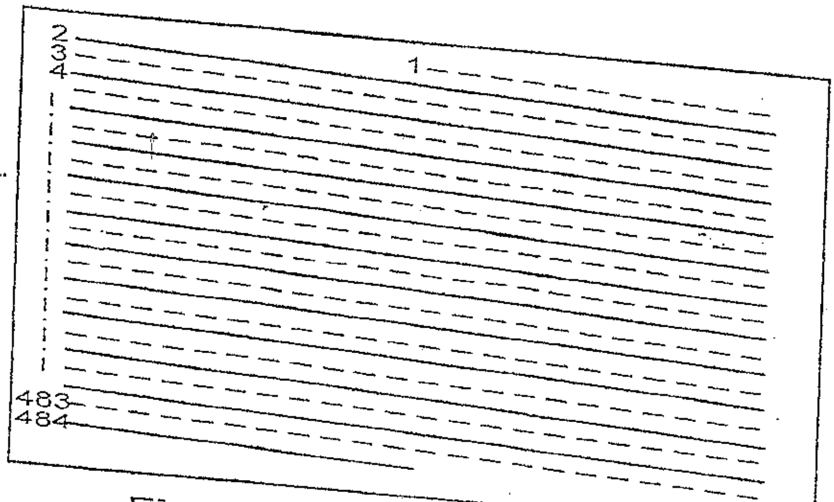


Figure 24: Interlaced Scanning

would no longer be apparent, but the video bandwidth would have to be doubled which would be very costly. Instead, the process of *interlaced scanning* is used to trick the human eye into believing it is seeing 60 pictures per second.

Figure 24 illustrates the process of interlaced scanning. The first set of lines (even field) is traced in $\frac{1}{60}$ s, and then the second set of lines (odd field) that make up the full scene (485 lines total) is interleaved between the first lines in the next $\frac{1}{60}$ s. Therefore lines 2, 4, 6, etc., occur during the even field, with lines 1, 3, 5, etc., interleaved between the even numbered lines in the odd field. The field frequency is thus 60 Hz, with a frame frequency of 30 Hz.

Horizontal Synchronization

To accommodate the 525 scanning lines (485 visible) every $\frac{1}{30}$ s, the transmitter must send a synchronization (sync) pulse between every line of video signal so that perfect transmitter-receiver synchronization is maintained. The detail of these pulses is

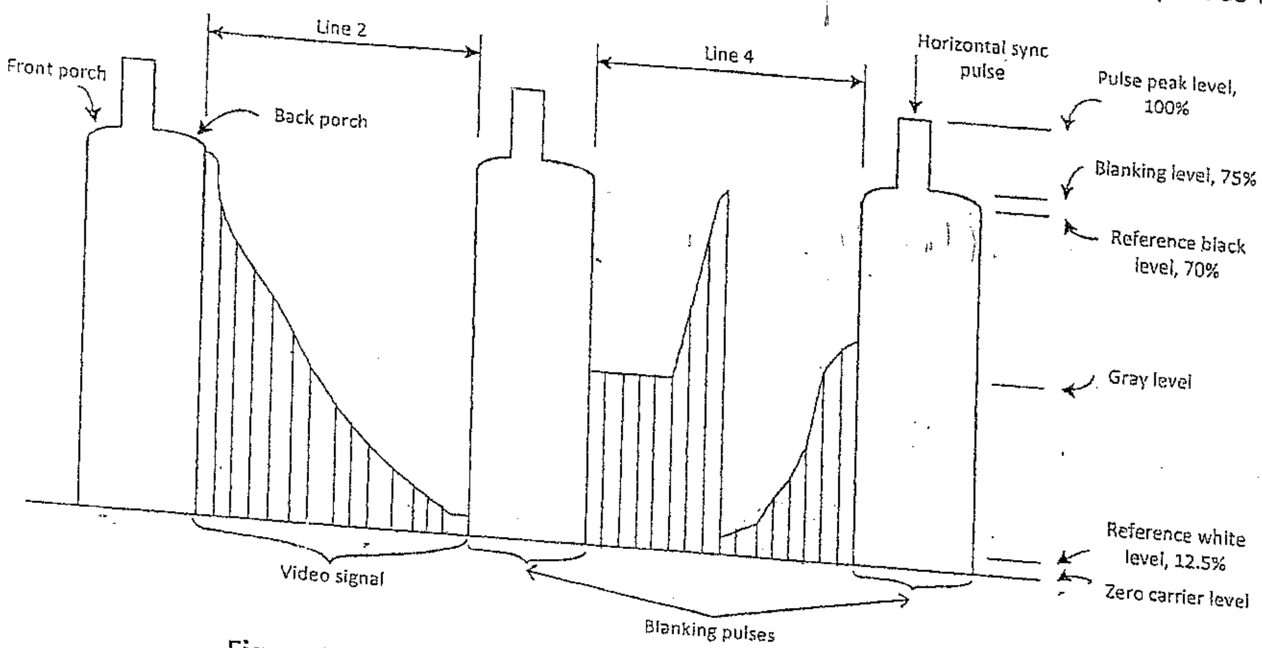


Figure 25: Horizontal blanking and sync pulses

shown in Figure 25. Three horizontal sync pulses are shown along with video signal for two lines. The sync pulses are superimposed on top of the horizontal blanking pulses as shown in the figure. The *blanking pulse* is a strong enough signal that is sent alongside the video signal during transmission so that the electron beam retrace at the receiver is "blacked" out and thus invisible to the viewer. The interval before the horizontal sync pulse appears on the blanking pulse is termed the *front porch*, while the interval after the end of the sync pulse, but before the end of the blanking pulse, is called the *back porch*.

The two lines of video picture signal shown in Figure 25, can be described as follows:

Line 2: It starts out nearly full black at the left-hand side and gradually lightens to full white at the right-hand side.

Line 4: It starts out medium gray and stays there until one-third of the way over, and it gradually becomes black at the picture center. It suddenly shifts to white and gradually turns darker gray at the right-hand side.

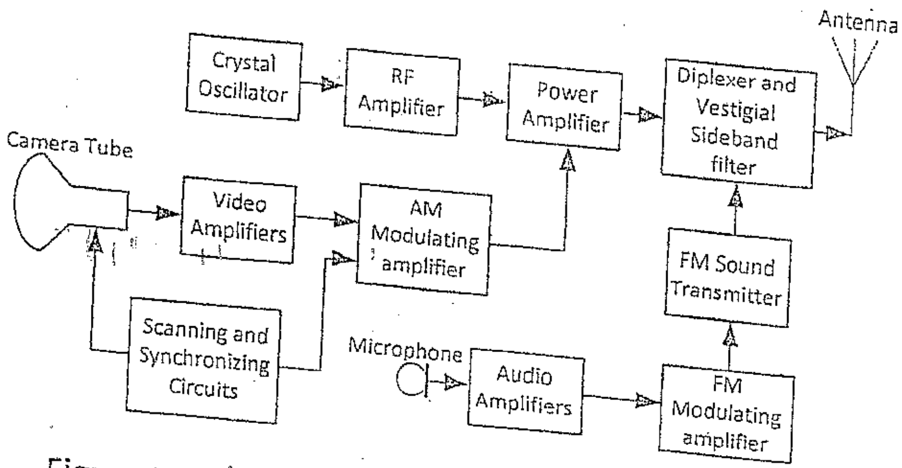


Figure 26: Black and white (Monochrome) TV transmitter

Colour Television

The introduction of colour television presents a much greater sophistication into the television engineering. One area of challenge was to make the black and white TV compatible with the colour TV, and this was successfully accomplished.

Compatibility of colour TV systems means that a colour TV system signal can be transmitted and be received by a black and white TV system. Also a black and white TV signal can be transmitted to be displayed in the colour TV system.

Conditions for compatibility: For compatibility to take place, the colour television system must:

1. Transmit, and be capable of receiving a luminance signal which is either identical to a monochrome transmission, or easily converted to it.
2. Use the same 6 – MHz bandwidth as monochrome TV.
3. Transmit the chroma information in such a way that it is sufficient for adequate colour reproduction, but easy to ignore by a monochrome receiver in such a way that no interference is caused by it.

Principles of Colour Television

At the colour TV transmitter the scene to be televised is actually scanned by three separate pickup tubes in the camera, each being sensitive only to one of the three primary colours, red, blue and green. Since the various combinations of these three colours can be mixed to form any colour to which the human eye is sensitive, an electrical representation of a complete colour scene is possible. The three colour cameras scan the scene in unison, with the red, green and blue colour content separated into three different signals. At the receiver these three separate signals are made to illuminate groups of red, green and blue phosphor dots (called *triads*), and the original scene is reproduced in colour.

After generation these three separate colour signals are fed into the transmitter signal processing circuits (*matrix*) and create the Y, or *luminance* signal and *chroma*, or colour signals I and Q. The Y signal contains just the right proportion of red, blue and green such that it creates a normal black-and-white picture. It modulates the video carrier just as does the signal from a single black-and-white camera with a 4-MHz bandwidth. The chroma signals, I and Q, are used to phase modulate the 3.58-MHz colour subcarrier, which then *interleaves* their colour information in the gap left by the luminance Y signal's sidebands. This modulation by I and Q signals is accomplished by a balanced modulator, thus suppressing the 3.58-MHz subcarrier, as it would cause interference at the receiver.

At the receiver, a monochrome set will simply detect the Y signal and thus present a normal black-and-white rendition of a colour picture. The chroma signals (I and Q) cannot be detected in a monochrome set because their 3.58-MHz subcarrier has been suppressed and is not present in the received signal. Thus a colour TV set must have a means to generate and reinject the 3.58-MHz subcarrier to enable detection of the I and Q signals.

Chroma effect equation

ke

$$Y = 0.30R + 0.59G + 0.11B$$

$$I = 0.60R - 0.28G - 0.32B$$

$$Q = 0.21R - 0.52G + 0.31B$$

Spectrum analyzer

Additional Notes on TV System

Persistence of Vision: Persistence of vision is the ability for the human eye to retain light in a fraction of time, when the scene changes from light to darkness, or the ability for the eye to continue to see darkness in a moment when the scene changes from darkness to light.

Table: Standards of Major Television Systems

Standard	American System	European System
1. Number of lines per frame	525	625
2. Number of frames per second	30	25
3. Field frequency, Hz	60	50
4. Line frequency, Hz	15,750	15,625
5. Video bandwidth, MHz	4.2	5

Exercise: Using the above table solve this problem:
The line frequency of a standard TV is given as 18250 Hz. If the number of lines per frame of picture is 625, what is the error associated with the line frequency when the number of picture frames per second is 25 and why is it so? Which country is the TV from? Britain Mention the name of its monochrome and colour TV systems. — PAL
The American colour TV system is called what? — NTSC

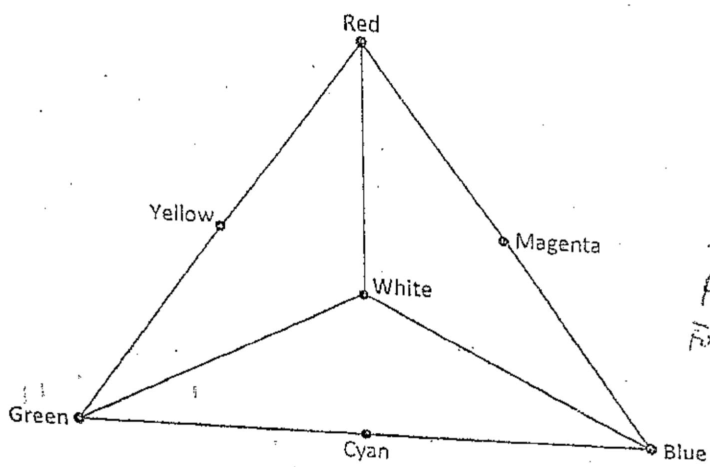
625 x 25 = 15625
18250 - 15625 = 2625

Synchronization in TV systems: Synchronization in television is a process whereby the speed of the transmitter scanning must be exactly duplicated by the receiver scanning process in order to avoid distortion in the receiver output. For example, when the TV camera starts line 1, the receiver must also start line 1 on the CRT output display.

<u>Aspect Ratio:</u> This is the ratio of the width of the picture to the height of it, and it is standardized to 4:3. It is the most pleasing picture orientation to the human eye.

Colour Mixing in TV Engineering: The type of colour mixing used in TV engineering is called <u>additive</u> colour mixing, while that of pigments is called <u>subtractive</u> colour mixing.

The Colour Triangle:



Additive Colour Triangle

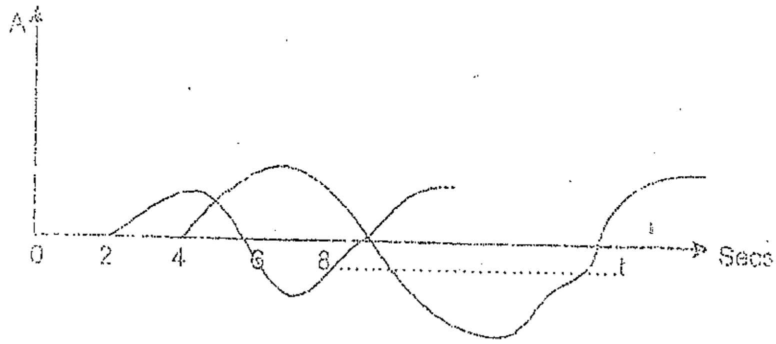
Radio & Television
For Test

Holly 6-8780

DIGITAL COMMUNICATION II (EEE 439)

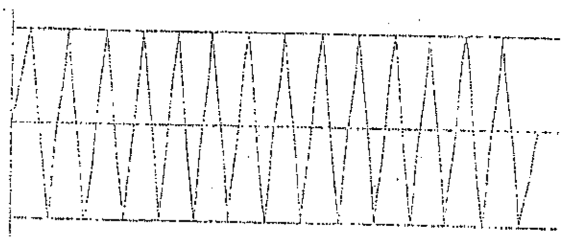
TEST 1C MODEL ANSWERS: HND II 2013/2014 SESSION DATE: 6/06/2014

)(a)(i)

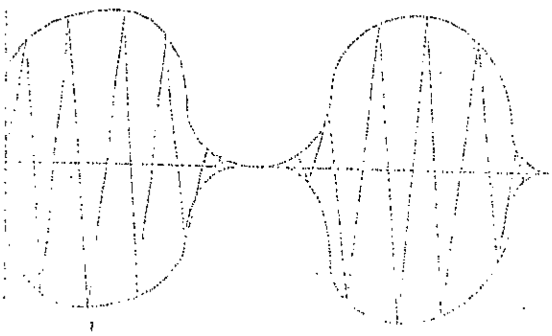


Given $f_s = 5 \sin \omega t$, $f_{s1} = 5 \sin \omega(t+2)$ secs; $f_{s2} = 5 \sin \omega(t+4)$ secs

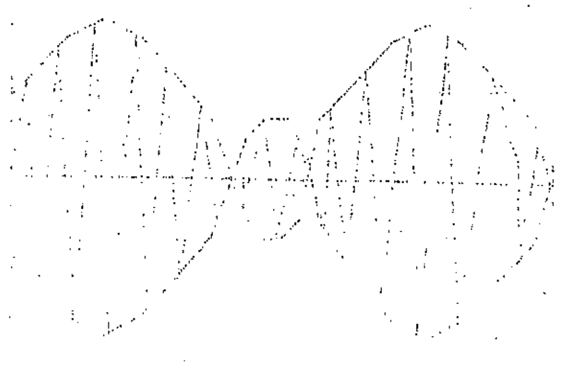
(ii)



0% Modulation



100% Modulation



120% Modulation